WORKED SOLUTION

STEP I Specimen paper, Question 4

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q4

The key idea

Whenever a square root of a quadratic appears, complete the square first. Here the quadratic under the root becomes $(x+1)^2-3$, and the factor outside the root is also $x+1$, so shifting to $u=x+1$ leaves the tidy form $\dfrac{1}{u\sqrt{u^2-3}}$. That shape (a variable times the root of its square minus a constant) is exactly what a secant substitution, or equally a reciprocal substitution $t=1/u$, is built to handle. Either one also turns the infinite range into a finite one.

Solution

Completing the square

We have

$$x^2+2x-2=(x+1)^2-3,$$

which is positive for $x\geq 1$, since then $(x+1)^2\geq 4>3$, so the integrand is defined throughout $[1,\infty)$. Put $u=x+1$, so $\mathrm{d}u=\mathrm{d}x$, $u=2$ when $x=1$, and $u\to\infty$ as $x\to\infty$:

$$I=\int_1^\infty \frac{\mathrm{d}x}{(x+1)\sqrt{x^2+2x-2}}=\int_2^\infty \frac{\mathrm{d}u}{u\sqrt{u^2-3}}.$$

For large u the integrand behaves like $1/u^2$, so the integral converges.

The secant substitution

Let $u=\sqrt{3}\sec\theta$ with $0\leq\theta<\frac{\pi}{2}$. Then

$$\mathrm{d}u=\sqrt{3}\sec\theta\tan\theta\,\mathrm{d}\theta,\qquad \sqrt{u^2-3}=\sqrt{3(\sec^2\theta-1)}=\sqrt{3}\tan\theta,$$

where we have used $\tan\theta\geq 0$ on this range to drop the modulus.

Ask yourself What are the new limits?

At $u = 2$ we need $\sec\theta = \frac{2}{\sqrt{3}}$, that is $\cos\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{\pi}{6}$. As $u \to \infty$, $\sec\theta \to \infty$, so $\theta \to \frac{\pi}{2}$ from below. The substitution is increasing on $[\frac{\pi}{6}, \frac{\pi}{2})$, so the order of the limits is preserved.

Everything now cancels:

$$\begin{aligned} I &= \int_{\pi/6}^{\pi/2} \frac{\sqrt{3}\sec\theta\tan\theta}{\sqrt{3}\sec\theta\cdot\sqrt{3}\tan\theta}\,\mathrm{d}\theta \\ &= \frac{1}{\sqrt{3}}\int_{\pi/6}^{\pi/2}\mathrm{d}\theta = \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}}. \end{aligned}$$

So

$$\int_1^\infty \frac{\mathrm{d}x}{(x+1)\sqrt{x^2+2x-2}} = \frac{\pi}{3\sqrt{3}} = \frac{\sqrt{3}\,\pi}{9}.$$

An alternative: the reciprocal substitution

If secants don't come to mind, $t = 1/u$ works just as well and makes the range finite at once. With $u = 1/t$, $\mathrm{d}u = -\mathrm{d}t/t^2$, and for $t > 0$

$$u\sqrt{u^2-3} = \frac{1}{t}\cdot\frac{\sqrt{1-3t^2}}{t} = \frac{\sqrt{1-3t^2}}{t^2}.$$

The limits $u = 2$ and $u \to \infty$ become $t = \frac{1}{2}$ and $t \to 0$, and the minus sign reverses them:

$$I = \int_0^{1/2}\frac{\mathrm{d}t}{\sqrt{1-3t^2}} = \left[\frac{1}{\sqrt{3}}\arcsin(\sqrt{3}\,t)\right]_0^{1/2} = \frac{1}{\sqrt{3}}\arcsin\frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}\cdot\frac{\pi}{3},$$

in agreement with the first method.

Checks

- Differentiate the antiderivative $F(x) = \frac{1}{\sqrt{3}}\operatorname{arcsec}\left(\frac{x+1}{\sqrt{3}}\right)$ and you recover the integrand; then $F(\infty) - F(1) = \frac{1}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right)$.
- A rough size check: on $[1,\infty)$ the integrand is a little bigger than $1/(x+1)^2$, whose integral is $\frac{1}{2}$. Our answer is $\pi/(3\sqrt{3}) \approx 0.6046$, slightly above $\frac{1}{2}$, as it should be.
- Numerical quadrature of the original integral (splitting the range and removing the infinite tail by $x = 1/w$) gives 0.604600, matching $\pi/(3\sqrt{3}) = 0.6045998\ldots$.

Common slips

- Forgetting to change the limits: the lower limit is $\theta = \frac{\pi}{6}$, not 0; using 0 gives the wrong answer $\pi/(2\sqrt{3})$.

- Writing $\sqrt{\sec^2\theta - 1} = \tan\theta$ without checking the sign of $\tan\theta$ on the range used.
- Leaving the answer as $\frac{1}{\sqrt{3}}\operatorname{arcsec}(\ldots)$ evaluated "at infinity" without taking the limit properly.

Beyond STEP

This is an improper integral: strictly, it means $\lim_{X\to\infty}\int_1^X$, and a first-year analysis course makes precise when such limits exist (comparison with $\int^\infty u^{-2}\,\mathrm{d}u$ is the standard test). The substitutions here are also the start of the general theory of integrating $R\left(x, \sqrt{ax^2+bx+c}\right)$ for rational R, which can always be reduced to integrating a rational function. For improper integrals and the comparison test, see Abbott, *Understanding Analysis*.

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